Home Energy Tracker Publication Update Audit

Evidence-based revision package for the IEEE-style manuscript

1. Scope and source control

The attached source manuscript version-1.1aHome_Energy_Tracker-publication_updated.docx was used because the filename with “(2)” named in the instruction was not present. The original was not overwritten. This audit accompanies the updated copy and documents the evidence status of substantive changes.

2. Content-audit summary

Area	Status	Action	Reason/evidence
Title, author block, references	Retained	No content redesign	Unchanged to preserve manuscript identity; 24 reference entries remain cited.
Abstract and contributions	Revised	Current baseline/REFIT distinction	Live baseline health/prediction added; REFIT declared disabled; historical metrics labelled historical.
Architecture and methodology	Revised	Correct worker/data-flow wording	FastAPI-hosted KafkaInferenceWorker, device-bound token, imputation, downstream credentials, and optional fallback clarified.
Figures 1–2	Replaced in place	Update active and disabled branches	Figures retain numbering, placement, and navy/light-blue box language while adding baseline, imputation, Kafka topics, and disabled REFIT.
Tables I–III	Revised	Historical/current separation	The old numeric results are retained under their conditions; current live runtime rows added without treating them as accuracy measurements.
Results, limitations, conclusion	Revised	Bound claims	New inference is functional-only; security, observability, and REFIT limitations stated explicitly.

3. Section-by-section change summary

Section	Substantive update
Abstract	Adds the active artifact identity, current live functional prediction, four expected imputations, and disabled REFIT state; historical metrics retained as historical.
I–II	Retains integration research gap; adds baseline/REFIT distinction and device-bound admission boundary without claiming a new ML algorithm.
III-A	Corrects the ML description: KafkaInferenceWorker resides in the FastAPI ML service, consumes energy-usage, produces anomaly-results, and routes repeated failures to the failure topic.
III-B	Documents baseline feature construction, training-median imputation, optional Ollama deterministic fallback, and REFIT isolation.
III-C	Separates historical artifact/E2E evidence from current runtime verification and disabled experimental REFIT.
IV	Labels older synthetic E2E/k6/outage records as historical and adds 2026-10-11 availability, authorization/event-path, ML functional, and separate regression evidence.
V	Adds current runtime rows to Table III and makes the discussion distinguish functional inference from accuracy/latency/full post-restart event-to-alert proof.
VI–VII	Adds disabled REFIT, static-shared-secret, no fresh all-target snapshot, and no browser acceptance-test limitations; preserves non-production boundary.

4. Figure and table update summary

Identifier	Update
Fig. 1	Retained in-place architecture figure; now shows the active baseline Random Forest and Kafka worker, device-bound token, anomaly-results/energy-alerts flow, optional Ollama fallback, Prometheus/Grafana, and a dashed disabled REFIT branch.
Fig. 2	Retained in-place workflow figure; separates historical artifact preparation, current runtime feature imputation/inference, evidence boundary, and disabled REFIT route.
Table I	Revised component/data-flow descriptions; identifies the baseline version and current REFIT-disabled boundary.
Table II	Retains host/stack facts while adding the active artifact/threshold and historical versus current evidence conditions.
Table III	Relabels old rows as historical and adds current live baseline plus authorization/event-path rows. It does not convert a single prediction into a performance metric.

5. Claim, evidence, and status matrix

Claim	Evidence	Status
17-service local Compose architecture	docker-compose.yml and 2026-10-11 validation notes	Verified configuration/local deployment
Gateway/frontend/Keycloak/Prometheus/Grafana availability	2026-10-11 HTTP 200 checks	Current verified availability only
Active baseline model	model_metadata.json; live /health	Current verified: random_forest-csv-20261003T205417Z
Live baseline functional inference	2026-10-11 /predict result	Current verified functional check only
Four missing runtime contextual features	main.py feature preparation and response	Current verified imputation behavior
REFIT controlled branch	Compose flags and live health	Disabled; no live/empirical result used
Historical RF metrics	model_metadata.json and historical evaluation files	Retained as historical supplied-CSV result
Historical k6 and Prometheus values	dated historical evidence	Retained with dates/conditions; not repeated
Authorization/event path	2026-10-11 live matrix and 0.37+0.19 kWh run	Current functional verification
Source-level regressions	46 Java tests, 18 local Python tests, frontend lint/build	Separate test records; not a composite reliability score

6. Retained historical results

- Baseline Random Forest held-out CSV metrics and Logistic Regression comparison, explicitly identified as historical artifact evaluation on the repository-supplied labelled CSV.
- Synthetic controlled E2E analytics, threshold alert/Mailpit, gateway-status, deduplication, Prometheus target, k6, and selected outage observations, each retained with the original bounded conditions.
- No historical measurement has been described as a fresh 2026-10-11 rerun.

7. Revised or removed obsolete/misleading claims

- Revised generic ML wording to the actual FastAPI-hosted KafkaInferenceWorker and baseline artifact flow.
- Removed any implication that REFIT is deployed, provides live predictions, consumes Wokwi traffic, or contributes final empirical metrics.
- Revised current-prediction wording so anomalyProbability = 0.0 is not presented as an accuracy, confidence, latency, or generalisation result.
- Revised observability wording: current Prometheus/Grafana HTTP availability is verified; the 8/8 target snapshot remains historical.
- Revised optional AI wording: current Overview/Saving Tips result was deterministic fallback, not verified live generative output.
- Revised security wording: current controls use local Keycloak/gateway/device-bound tokens/static internal secrets and are not production mTLS, workload identity, rotation, replay protection, or penetration-test evidence.

8. Unresolved or manual-verification items

- The provided manuscript filename differs from the filename named in the instruction; this update used the attached accessible source document.
- No physical-meter calibration, utility-bill agreement, consented household field data, or external generalisation experiment is available.
- CSV provenance, licence, label-generation procedure, household/device identity, and unit semantics remain undocumented.
- The optional REFIT artifacts are unavailable to the current Docker build and its controlled benchmark must not be represented as deployed or real-anomaly performance.
- No fresh telemetry-to-Kafka-to-ML-to-alert completion, post-ML-restart all-target Prometheus sweep, browser role/refresh acceptance test, latency distribution, or production security assessment was captured.

9. Final formatting and consistency audit

- Preserved: title, authors, affiliations, corresponding-author designation, two-column section structure, section-break paragraphs, figure/table numbering and captions, reference numbering, and 24-entry reference list.
- Verified in the edited DOCX structure: 98 body paragraphs, 3 tables, 2 inline figures, and 12 document sections, matching the source structural counts.
- Terminology scan completed for misleading REFIT deployment, real-world accuracy, production-ready, and fully resilient claims; replacement text uses bounded, evidence-specific language.
- Page-level rendering was completed with Microsoft Word PDF export and Poppler rasterization; all seven manuscript pages and all three audit pages were visually inspected before delivery.